Design of Low-Current Thermionic Electron Gun

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Objectives

With increasing pollution, availability of clean water is reducing day by day. A novel solution to this problem, lies in using electron guns for water purification. The high energy from the electrons can be harnessed to break the bonds in impurity radicals. This precipitates the impurities, making their removal much easier. Computer simulations play a crucial role in determining the final electron gun parameters.

Methodology

COMSOL Multiphysics

Charged Particle Tracing

Specifying beam current, no of macro particles, gun geometry, potential.

Space Charge Limited Emission

Plotting Graphs(Particle trajectory, Trace Space)

Pierce Method of Gun Design

The Child Law for space charge limited emission is

$$J_o = \frac{4\,\epsilon_0}{9}\sqrt{\frac{2Ze}{m_0}}\frac{V_o^{3/2}}{d^2}$$

The Electric potential in the gap follows the potential,

A complex analytical function will also be a solution of the Laplace equation and real part of that function will be a valid expression for V. Expressing the complex function in Euler's notation and applying the boundary conditions, we have $\frac{4\theta}{3} = \frac{\pi}{2}$ or $\theta = 67.5\ degree.$

Finite Element Method Simulation

Solving Poisson Equation

$$\nabla^2 V = \frac{\varphi}{\epsilon 0}$$

Finding **E** from φ

Advancing the particle orbits

$$\frac{\partial m_e \boldsymbol{v}}{\partial t} = \boldsymbol{F}$$

Reassigning space charge at all mesh points

Re-solving Poisson Equation(including space charge effects)

Results

We have obtained the particle trajectory for the 67.5-degree pierce angle, the trace space plots are shown below. The beam waist position gets closer to the cathode.

Figure 1: YY' Trace Space

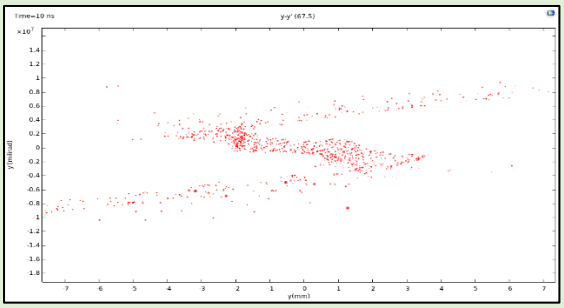


Figure 2: X-X' Trace Space

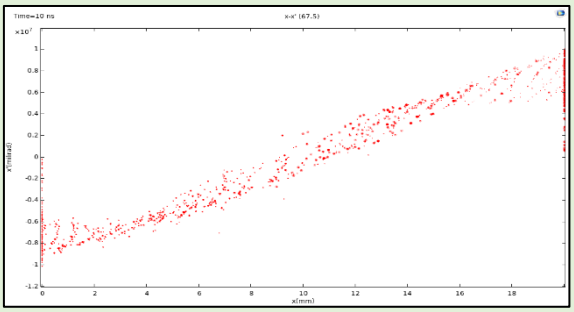
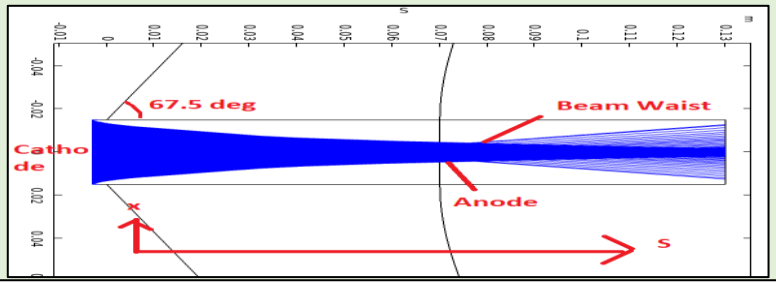


Figure 3: Particle Trajectories (67.5 degree)



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